Monte Carlo investigation of critical behavior in the two-dimensional Ising model with impurities and external magnetic fields

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ABSTRACT

Thermal phase transitions in the two-dimensional Ising model on square lattices were investigated using Metropolis Monte Carlo simulations with periodic boundary conditions to compute temperature-dependent energy, magnetization, specific heat, and susceptibility. Analysis of specific heat peaks for different lattice sizes yielded an estimated critical temperature, $T_c \approx 2.28\ J/k_B$, in close agreement with the exact value $T_c \approx 2.27\ J/k_B$. Static impurities at low and high concentrations were simulated, illustrating the effect of impurities on observables. Simulated annealing was employed, demonstrating fragmented magnetic domain formation and convergence to global energy minimum. The effect of weak external magnetic fields was examined through measurements of magnetization and susceptibility, revealing that external field shifts the critical temperature and affects the bias of magnetization. Scaling collapse of susceptibility data confirmed the two-dimensional Ising universality class and validated known critical exponents. These results demonstrate the effectiveness of Monte Carlo methods for capturing phase-transition phenomena in spin systems and illustrate the effects of impurities and external fields on critical properties.

I. INTRODUCTION

The Ising model, originally proposed by Ernst Ising in 1925 and solved in two dimensions by Lars Onsager in 1944, provides a simple foundation for analyzing phase transitions and critical phenomena in spin systems[1]. It consists of a lattice of discrete spins that interact with their nearest neighbors and can take values of ± 1. Despite its simplicity, the model captures essential features of ferromagnetic behavior, including magnetization and the influence of thermal fluctuations near the critical temperature.

This study used the two-dimensional Ising model to investigate thermal phase transitions using Monte Carlo simulations with the Metropolis algorithm. Square lattices with periodic boundary conditions were used to minimize boundary effects and better approximate thermodynamic behavior. The temperature dependence of key physical observables; energy, magnetization and specific heat were examined in the absence of an external magnetic field at different temperatures.

The extentions were then considered. First, static impurities were introduced to investigate their on observables, mimicking point defects in real magnetic materials. Simulated annealing was applied to explore low energy configurations in these disordered systems, reflecting thermal annealing processes used in material processing. Secondly, weak magnetic fields were applied to see the effect on the spin system, to explore more realistic operating conditions in magnetic technologies. This allowed for calculation of susceptibility and confirming critical behavior in realistic magnetic systems.

II. THEORY

The Ising Model

The Ising model is an important model in statistical physics, used to study phase transitions in magnetic systems[1]. In the two-dimensional version, spins $s_i = \pm 1$ are arranged on a square lattice, where each spin interacts only with its nearest neighbors. The Hamiltonian of a two dimensional Ising system is defined as

$$E = -J \sum_{\langle i,j \rangle} s_i s_j - \mu H \sum_i s_i \tag{1}$$

where J is the exchange constant (positive for ferromagnetic interactions), μ is the magnetic moment, H is the external magnetic field, and $\langle i, j \rangle$ denotes summation over nearest-neighbor pairs[2]. When $H = 0$, the system can undergo a phase transition at a critical temperature of $T_c = 2.27\ J/k_B$[3]. In this study, J, μ and k_B were set equal to 1, for measuring temperature in units of J/k_B and magnetic field in units of J/μ. This effectively makes T and H dimensionless.

Monte Carlo Simulation and the Metropolis Algorithm

To simulate the thermal behavior of the system, the Metropolis Monte Carlo algorithm was implemented[4]. In each Monte Carlo sweep, spins are randomly selected and flipped with the probability:

$$P = \min\left(1, e^{-\Delta E/k_B T}\right) \tag{2}$$

where ΔE is the change in energy due to the spin flip, k_B is the Boltzmann constant, and T is the temperature. This means that if $\Delta E \leq 0$ then the flip is accepted unconditionally, otherwise it flips with probability $p = e^{-\Delta E/k_B T}$. After sufficient sweeps, the system reaches equilibrium and physical observables can be averaged over generated states.

Thermodynamic Properties

Thermodynamic observables are computed by averaging over microstates. These quantities reveal how the system responds to thermal fluctuations and time.

The total energy is calculated using the Hamiltonian in Eq. 1. It serves as a baseline for evaluating thermal fluctuations and specific heat. The magnetization was calculated using:

$$M = \sum_i s_i \tag{3}$$

summing over all the spins. The specific heat, C, and susceptibility, χ, were computed using[5]:

$$C = \frac{\Delta E^2}{k_B T^2} \quad (4) \qquad \chi = \frac{\Delta M^2}{k_B T} \quad (5)$$

where $\Delta X = \langle X^2 \rangle - \langle X \rangle^2$.

Critical behavior

Near the critical temperature, T_c, the two-dimensional Ising model exhibits singular behavior in its thermodynamic observables. Defining the reduced temperature:

$$t = |T - T_c|, \tag{6}$$

The specific heat near the critical temperature is characterized by the exponent α[6]:

$$C \sim \frac{1}{|T - T_c|^{\alpha}} \tag{7}$$

For the two-dimensional square-lattice Ising model, $\alpha = 0$, which implies a logarithmic divergence of the specific heat, a horizontal line near T_c. However, at higher t, the ordinary thermal effects will dominate, causing C to deviate from Eq. 7.

Other thermodynamic observables display their own critical behavior at $t \to 0$. Below T_c the magnetization becomes more ordered and above T_c the system becomes more disordered[3]. The magnetic susceptibility will peak at T_c, reflecting the system's high responsiveness to an applied magnetic field.

Impurities and Simulated Annealing

To mimic real material imperfections, impurities were introduced as spins that cannot flip. Impurities affect the energy and magnetization of the system, by creating fragmented magnetic domains[7]. This creates an energy landscape with many energy minimums, which affects the specific heat and critical temperature. Simulated annealing is a method for finding low-energy configurations by gradually lowering the temperature. This allows the system to escape local minima and approach a global minimum energy state, which reveals how the impurities affect the lowest energy magnetic domain[8].

Scaling and Data Collapse

To study critical behavior in the presence of a magnetic field, scaling theory is used. The magnetic susceptibility follows the scaling relation, taken from the exam paper.

$$\chi(t,h) = |t|^{-\gamma} g_{\pm}\left(\frac{h}{|t|^{\beta\delta}}\right), \tag{8}$$

where $t = \frac{T - T_c}{T_c}$ is the reduced temperature, $h = \frac{\mu H}{J}$ is the reduced magnetic field and $g_{\pm}$ are scaling functions above and below T_c. By plotting $\chi|t|^{\gamma}$ against $h/|t|^{\beta\delta}$, results for different T and H will collapse onto universal curves. The critical exponents for the 2D Ising model are $\gamma = 7/4$, $\beta = 1/8$, and $\delta = 15$[9].

III. METHODOLOGY

Monte Carlo Simulations

The two-dimensional Ising model was simulated using the Metropolis Monte Carlo method. A square lattice of size $L \times L$ was created, where each lattice site carries a random spin $s_i = \pm 1$. Periodic boundary conditions were imposed to minimize edge effects.

At each Monte Carlo sweep, a randomly chosen spin was considered for flipping. The energy change ΔE associated with the flip was computed, and the spin was flipped with a probability, calculated from Eq. 2. This algorithm ensures detailed balance and samples the Boltzmann distribution.

For all simulations, equilibrium was reached by doing $\approx$ 2000 sweeps, then calculating observables over $\approx 5000 - 10000$ sweeps, depending on the simulation. This was done to ensure the system had fully equilibrated before measurements and to collect a sufficiently large set of largely samples for reliable estimation of thermodynamic quantities.

Calculation of Physical Quantities

For each simulation, depending on the experiment, the energy, magnetization, specific heat and magnetic susceptibility were calculated using formulas from the theory. For scaling, these values were divided by L^2 to get the values per spin.

Clean System Simulations

To analyze the thermal behavior and identify the critical temperature of the Ising model, simulations were performed on square lattices of varying size ($L = 10$, 20, 30, and 40) without impurities or external fields. Observables such as average energy, magnetization, and specific heat were computed over a range of temperatures. Particular focus was given to the temperature region around the critical point $T_c \approx 2.27$. The critical behavior of C was observed by plotting C against $|T - T_c|$ in logarithmic space, and using linear regression to find α.

Impurity Simulations

Impurities were introduced by randomly selecting a fraction p of spin sites to be non-flippable. These spins were initialized randomly like the rest but were excluded from Metropolis updates during the simulation. This mimics structural defects in real materials.

Simulations were performed for impurity concentrations $p = 0.02$, 0.1, and 0.25. The effect on thermodynamic observables was analyzed by comparing energy, magnetization, and specific heat as functions of temperature. The same fixed spins were used for all temperatures, as to ensure any observed changes could only be attributed to thermal effects, and not variation in impurity configuration.

Additionally, simulated annealing was used to investigate ground-state configurations in the presence of impurities. Starting from a high temperature, the system was slowly cooled from higher temperatures to a near-zero temperature. The spin configuration with the lowest energy was saved for each impurity concentration, from $T = 5$ to $T = 0.1$. The lowest energy states were analyzed to study how impurities disrupted magnetization and energy in the lowest energy state.

External Field Simulations

To study the magnetic susceptibility scaling, an external magnetic field H was introduced. The Hamiltonian was modified to Eq. 1. First the magnetization at $T = 2.3$ and $H = 0.01$ was measured over time to observe the system's reaction to a magnetic field. Then the effect of different magnetic field ($H = 0.01$, $H = 0.03$ and $H = 0.05$) was observed over different temperatures. The susceptibility was also calculated using Eq. 5. Lastly scaling collapse plots were produced by rescaling susceptibility and magnetic field near T_c, using Eq. 8. Here, plotting $\chi|t|^{\gamma}$ against $h/|t|^{\beta\delta}$.

Implementation Details

The simulations were implemented in Python using NumPy, Matplotlib, and Numba for performance optimization. Spin lattice updates and energy calculations were accelerated using Just-In-Time (JIT) compilation[10]. Data from simulations were saved and loaded as compressed npz files to reduce computational time during repeated analyses.

IV. RESULTS AND DISCUSSION

Behavior of a clean system

Fig. 1 shows the evolution of magnetization and energy over time for three different temperatures: $T = 2.1$, 2.3, and 2.5. At low temperature ($T = 2.1$), the system quickly reaches a stable, magnetized state. Near the critical temperature ($T = 2.3$), large fluctuations in magnetization are appear, indicating critical behavior. At higher temperature ($T = 2.5$), the system fluctuates around zero magnetization, consistent with a paramagnetic phase[11]. The energy curves similarly reflect thermal fluctuations, with larger fluctuations near T_c. This is expected for temperature near the phase transition[12].

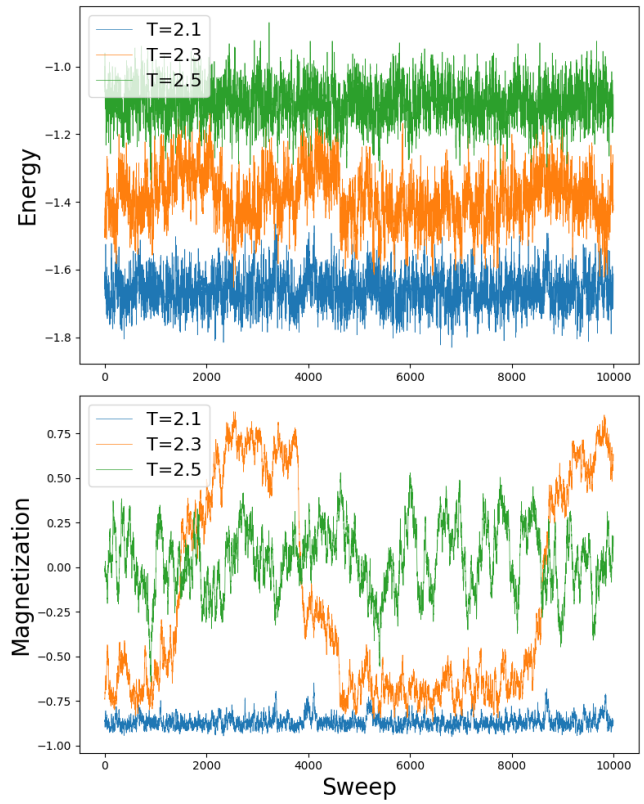

FIG. 1. Magnetization and energy over time.

In Fig. 2, the average energy and absolute magnetization are plotted as functions of temperature. The transition around the critical temperature, $T_c \approx 2.27$[3], is evident: magnetization drops sharply while energy increases steadily. Spin configuration snapshots further confirms this transition. At $T = 1.7$ the snapshot shows large ordered domain, while near T_c, $T = 2.3$, shows fragmented magnetic domains. At $T = 3.0$ the system demonstrates a fully disordered system. This progression is characteristic of a magnetic material near critical temperature[12].

To estimate the critical temperature, specific heat was computed for lattice sizes $L = 10$, 20, 30, and 40. As shown in Fig. 3, the specific heat exhibits sharp peaks, demonstrating the phase transition. The location of these peaks was used to estimate the critical temperature. The critical temperature was calculated to be $T_c = 2.28$, very close to the theoretical value of $T_c = 2.27$[3]. These results validate the reliability of the simulation approach and the accuracy of the Metropolis algorithm for capturing critical behavior.

Plots of specific heat against the reduced temperature in logarithmic space, Fig. 4, show that the curves for different system sizes follow the same behavior. At tempera-

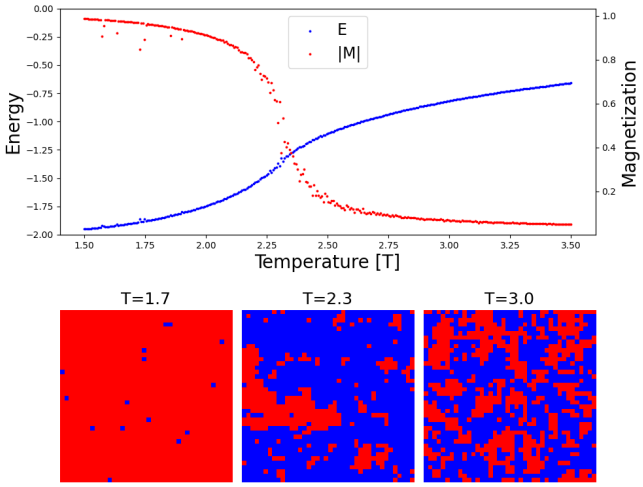

FIG. 2. Average magnetization and energy over temperature, showing snapshots at different temperatures.

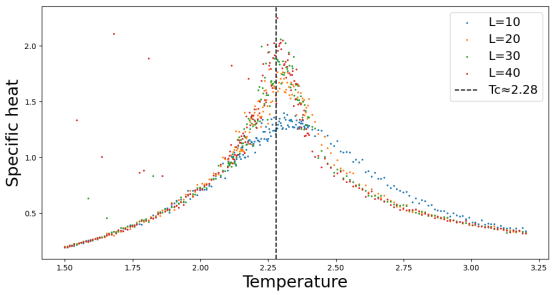

FIG. 3. Specific heat over temperature for different sized lattices.

tures close to the critical temperature, the systems collapse onto a nearly straight-lined behavior, consistent with theoretical expectations[6]. Extracting α using Eq. 7 yields $\alpha \approx -0.01, 0.02, 0.02$ and 0.09, consistent with expectations of $\alpha = 0$. The inaccuracies stems from there being fewer samples near T_c where larger fluctuations dominate, however the trend is easily observed. At larger reduced temperatures, the curves turn downward and no longer follows the expected power-law trend. This is because, as the temperature moves further from T_c, the critical fluctuations fade and ordinary thermal effects dominate.

Effect of Impurities and simulated annealing

The impact of impurities was studied by introducing fixed (non-flippable) spins. Fig. 5 shows that increasing impurity concentration ($p = 0.02, 0.1, 0.25$) reduces magnetization and specific heat, while increasing system energy. This occurs because impurities hinder spin alignment, reducing the system's ability to form large magnetic domains. The peaks for specific heat are both less pronounced and placed at a lower temperature, demonstrating the expected effect of impurities on phase transition[13].

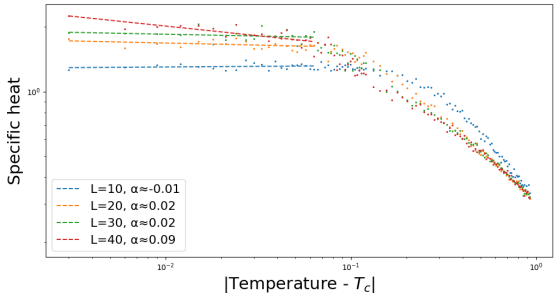

FIG. 4. Logarithmic plot of the specific heat, C, versus reduced temperature, $|T - T_c|$ for different system sizes.

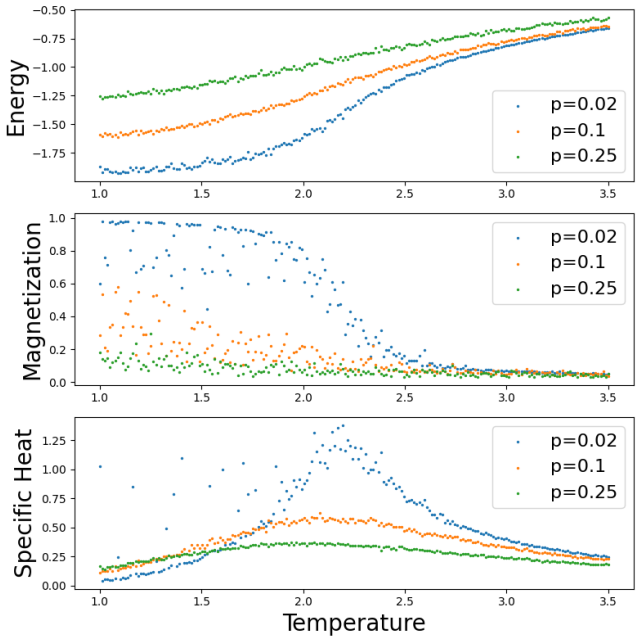

FIG. 5. Energy, magnetization and specific heat plotted against temperature, with different impurity concentrations.

Simulated annealing was applied to explore low-energy configurations in systems with impurities, by slowly reducing the temperature while simulating. As shown in Fig. 6, even with high impurity concentrations, the system was able to form partially ordered domains. In the presence of impurities, the lowest energy state demonstrates increased energy and reduced magnetization compared to the cleaner systems. The fixed impurity spins disrupt magnetic order and prevent the system from fully minimizing its energy, resulting in fragmented magnetic domains. These results demonstrate how local constraints imposed by disorder prevent the system from reaching a fully ordered ground state, despite slow cooling.

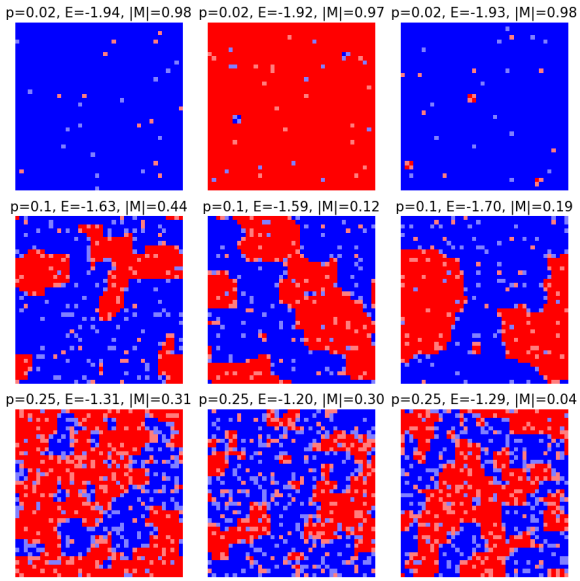

FIG. 6. Magnetic configuration in the lowest energy state after simulated annealing. Lighter colored sites are fixed, while darker are flippable.

Influence of External Magnetic Field

The influence of an external magnetic field was investigated at $T = 2.3$, just above the critical temperature. Fig. 7 shows that applying a small magnetic field ($H = 0.01$) biases the magnetization direction, breaking the spin-reversal symmetry. This results in persistent net magnetization even near criticality, suggesting increased T_c. The snapshots confirm the magnetic bias; all final configurations tend toward the field direction, even though fluctuations are still there.

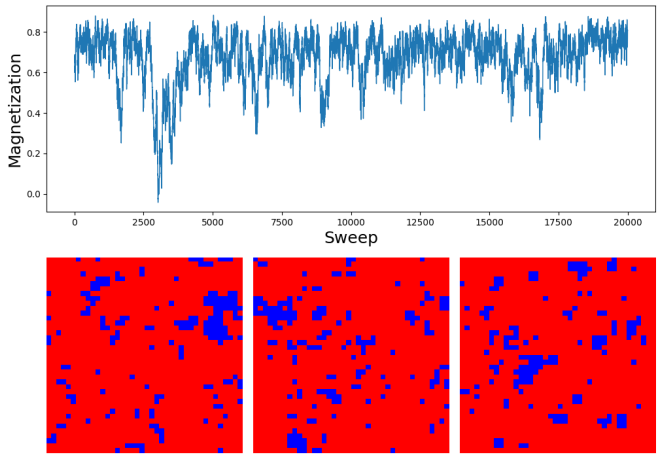

FIG. 7. Magnetization over time with $H = 0.01$, showing three snapshots.

Fig. 8 displays the temperature dependence of magnetization and susceptibility for three different external fields. As field strength, H, increases, the magnetization decreases more gradually with temperature, demonstrating the effect of the magnetic field stabilizing the magnetization. The susceptibility exhibits more pronounced peak for smaller H, since weaker fields allow the system to have larger fluctuations in spin alignment. This indicates that weaker fields show more critical-like behavior near the phase transition, while stronger fields suppress these fluctuations and smoothens the transition. The peak in susceptibility also shifts to higher temperatures, indicating that the critical temperature is shifted with an external magnetic field. This is consistent with theoretical expectations, as an external magnetic field breaks the symmetry of the Ising model, favoring the alignment in one direction, which delays the phase transition to higher temperatures[1].

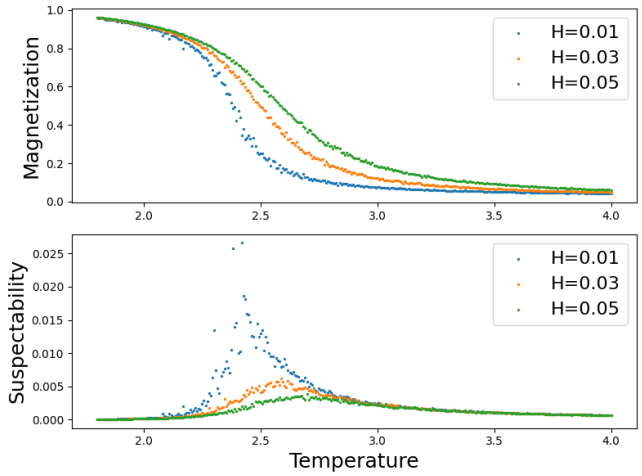

FIG. 8. Magnetization and susceptibility over temperature.

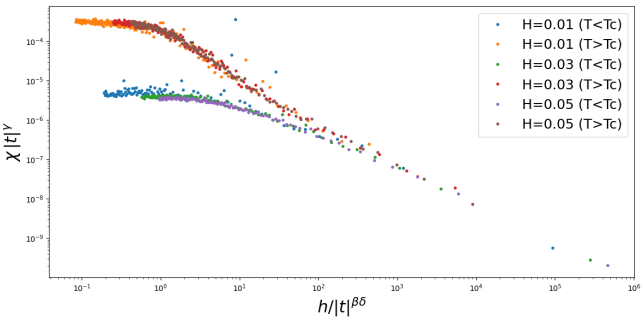

FIG. 9. Scaling collapse of susceptibility.

Finally, the susceptibility data were rescaled according to Eq. 8. with $t = (T - T_c)/T_c$. Fig. 9 shows that after rescaling, the data collapse onto two universal branches (above and below T_c), with H showing no difference in values. This confirms the validity of the critical exponents $\gamma = 7/4$, $\beta = 1/8$, and $\delta = 15$ for the 2D Ising universality class[9].

V. CONCLUSION

This study showed that Monte Carlo simulations of the two-dimensional Ising model can reproduce the characteristics of a thermal phase transition. Time-series of magnetization and energy at temperatures below, near, and above the critical point reveal rapid stabilization into an ordered state at low T, pronounced critical fluctuations near T_c, and paramagnetic behavior at high T. Locating the peak of the specific heat for various lattice sizes yielded an estimate of $T_c \approx 2.28$, in close agreement with the theoretical value $T_c \approx 2.27$. It was also observed that the critical behavior of the specific heat followed the expected characterized behavior.

Introducing static impurities at concentrations $p = 0.02$, 0.10, and 0.25 systematically suppresses the magnetization, broadens and lowers the specific heat peak, and raises the average energy, illustrating how fixed impurities affect the phase transition. Simulated annealing further highlights that, even under slow cooling, systems with impurities form several magnetic domains, confirming that local constraints hinder the approach to a fully ordered ground state.

Application of a weak external magnetic field ($H = 0.01$–0.05) biases the net magnetization and shifts the susceptibility peak to higher temperatures. Rescaling the susceptibility χ according to the reduced temperature $t = (T - T_c)/T_c$ and reduced magnetic field $h = \mu H/J$ produces a successful data collapse onto universal branches, validating the critical exponents $\gamma = \frac{7}{4}$, $\beta = \frac{1}{8}$, and $\delta = 15$ of the 2D Ising universality class.

This study confirms that the Monte Carlo approach captures the foundational behavior of idealized spin system. This model was successfully extended to explore the effect of static impurities and external magnetic fields. Future work could investigate larger lattices for more precise estimations, alternative disorder types and the effect of both magnetic field and impurities at the same time.

VI. ACKNOWLEDGMENTS

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VII. AI CONTRIBUTION

This report was made with the support of AI tools for structuring, grammar checking, and refining mathematical expressions. The AI did not generate original research results, perform data analysis, or influence the scientific conclusions presented in the study. AI tools were also used for help with optimizing original code.

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